Freeform Electrostatic Speakers

We present freeform interactive loudspeakers for creating spatial sound experiences from a variety of surfaces. Surround sound systems are widely used and consist of multiple electromagnetic speakers that create point sound sources within a space. Our proposed system creates directional sound and can be easily embedded into architecture, furniture and many everyday objects. We use electrostatic loudspeaker technology made from thin, flexible, lightweight and low cost materials and can be of different size and shape. In this paper we propose various configurations such as single speaker, speaker array and tangible speakers for creating playful and exciting interactions with spatial sounds.

Figure 1. Freeform Interactive Loudspeakers are flexible, lightweight and used in interactions with everyday objects. (a) The lightweight electrostatic loudspeakers are integrated in a *balloon*. (b) The electrostatic loudspeakers can be made into *large curved wall* surfaces. (c) Our *speaker table* accepts touches and weight on it. (d) *Tangible objects* are made speaker by equipping them with passive layer of conductive paper.

- Ishiguro, Y., Israr, A., Rothera, A., & Brockmeyer, E. (2014, November). Uminari: freeform interactive loudspeakers. In *Proceedings of the Ninth ACM International Conference on Interactive Tabletops and Surfaces* (pp. 55-64). [Link]

- Ishiguro, Y., Brockmeyer, E., Rothera, A., & Israr, A. (2014, October). Ubisonus: spatial freeform interactive speakers. In *Adjunct Proceedings of the 27th Annual ACM Symposium on User Interface Software and Technology* (pp. 67-68). [Link]

@https://www.youtube.com/watch?v=Q4ZU364QdoM